








Statistics of Financial Markets

Seminar 15: Systemic risk



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material

- **AI**: Artificial Intelligence
- **BAC**: Bank of America (ticker symbol)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNP**: Banque Nationale de Paris (BNP Paribas) — the French banking group BNP Paribas
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **COVID-19**: COronaVIrus Disease 2019
- **DBK**: Deutsche Bank (ticker symbol)
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **GS**: Goldman Sachs (ticker symbol)
- **HSBA**: HSBC Holdings (ticker symbol, London)
- **HSBC**: Hongkong and Shanghai Banking Corporation
- **ING**: Internationale Nederlanden Groep (the Dutch banking group ING)
- **INGA**: ING Groep (ticker symbol)
- **JPM**: JPMorgan Chase (ticker symbol)
- **LRMES**: Long-Run Marginal Expected Shortfall
- **MES**: Marginal Expected Shortfall
- **MS**: Morgan Stanley (ticker symbol)
- **SAN**: Banco Santander (ticker symbol)
- **SRISK**: Systemic RISK (capital shortfall in a crisis)
- **SVB**: Silicon Valley Bank
- **US**: United States
- **VAR**: Vector AutoRegression
- **WFC**: Wells Fargo and Company (ticker symbol)

Today's Question and Seminar Structure

- Question of the seminar
  - **which bank matters most for the stability of the system, and can we tell it from market prices?**
- Outputs to produce
  - the answers to the A exercises
  - the table or figure each B exercise asks for
  - a short argued answer to C1; the errors found in the AI answer of C2 and the corrected answer
  - the interpretation sentences each exercise asks for
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- The exercises, their types and their order: the Exercise Map (next slide)
- Nothing is handed in: the seminar is for practice; the solutions of the [Proposed] exercises are discussed in class
- Contact: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise Map

Task	Question	Type	Model
A1, A2	CoVaR 1% and ΔCoVaR from a quantile regression: JPMorgan Chase; Bank of America	Solved, Proposed	A1
A3, A4	MES and LRMES from 20 days of March 2020: JPMorgan Chase; Goldman Sachs	Solved, Proposed	A3
A5, A6	SRISK of one bank; SRISK of two banks and the aggregate SRISK	Solved, Proposed	A5
B1, B2	MES 5% and ΔCoVaR 1% with bootstrap intervals: US banks; European and Romanian banks	Solved, Proposed	B1
B3, B4	contagion or interdependence: US and European banks in 2008; European and Romanian banks in 2020	Solved, Proposed	B3
B5, B6	does MES before a crisis rank the losses in it? March 2020; March 2023	Solved, Proposed	B5
C1, C2	how connected are the Romanian and the euro-area banks? what is wrong in an AI answer?	Proposed	B1, A1

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model in the last column
- Part A: calculations on paper; Part B: real data, in the notebook, each ending with an interpretation question; Part C: open questions and the critique of an AI answer

Data Used

Series	Source	Price used	Period
6 US banks, 5 European banks	EODHD	adjusted close	2005–2026
Banca Transilvania (TLV), BRD	EODHD	adjusted close	2010–2026
S&P 500, Euro Stoxx 50, BET	EODHD	close	2005–2026

- US banks: JPMorgan Chase (JPM), Bank of America (BAC), Citigroup (C), Goldman Sachs (GS), Morgan Stanley (MS), Wells Fargo (WFC); European banks: Deutsche Bank (DBK), BNP Paribas (BNP), Santander (SAN), ING (INGA), HSBC (HSBA)
- Prices joined on common days first, then daily log returns in %; BVB days without trades removed; TLV without 30–31 May 2016; the last day is 18 September 2026
- Market of each bank: S&P 500 (US), Euro Stoxx 50 (Europe), BET (Romania); a bank portfolio: equal weights, rebalanced daily

Prerequisites for Today (1/14): Systemic Risk

- **Systemic risk**: the risk that the financial system as a whole stops working, with large costs for the real economy IMF, BIS & FSB (2009)
  - channels: direct contagion (exposures between banks), common exposures, fire sales, runs on deposits and funding
- The VaR of a bank (Chapter 10) looks at the bank alone; systemic measures look at the **joint tail** of a bank and the system
  - **CoVaR**: the system given that the bank is in distress Adrian & Brunnermeier (2016)
  - **MES** and **SRISK**: the bank given that the market crashes Acharya, Pedersen, Philippon & Richardson (2017); Brownlees & Engle (2017)
- Level α = the probability of the tail: VaR 1%, CoVaR 1%, MES 5%; losses are reported as positive numbers
  - a portfolio of N banks with equal weights, rebalanced daily: $r_t^p = 100 \ln\left(1 + \frac{1}{N}\sum_{i=1}^{N}(e^{r_{it}/100} - 1)\right)$
  - r_{it}: the log return of bank i on day t, in %; the bracket averages the simple returns

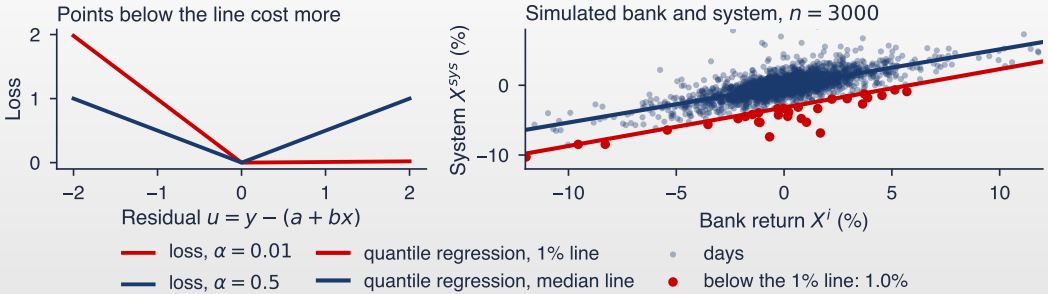
Prerequisites for Today (2/14): Quantile Regression

- $q_\alpha(X)$: the α-quantile, the value below which X falls with probability α; $\mathrm{VaR}_\alpha = -q_\alpha$ (Chapter 10)
  - $q_{50\%}$: the median, a typical day
- Linear **quantile regression** Koenker & Bassett (1978): $q_\alpha(Y \mid X = x) = a + bx$, the α-quantile of Y on the days when $X = x$
  - a: the intercept (constant term), b: the slope
- Estimation: choose a and b that minimise

$$\sum_t \rho_\alpha(y_t - a - bx_t), \qquad \rho_\alpha(u) = u\,(\alpha - 1\{u < 0\})$$

- Reading
  - $1\{u < 0\}$: 1 if the point is below the line, 0 otherwise
  - a point below the line costs $(1-\alpha)|u|$, one above it $\alpha|u|$; with $\alpha = 1\%$ the ratio is 99
  - result: about $100\alpha\%$ of the points lie below the fitted line

Prerequisites for Today (3/14): The Quantile Loss and the 1% Line



- Left: the loss $\rho_\alpha(u)$; for $\alpha = 0.5$ it is symmetric (the median), for $\alpha = 0.01$ almost all the cost is below the line
- Right: 3000 simulated days of a bank and of the system; exactly 1% of the points lie below the 1% line, which is steeper than the median line: the link is stronger in the tail

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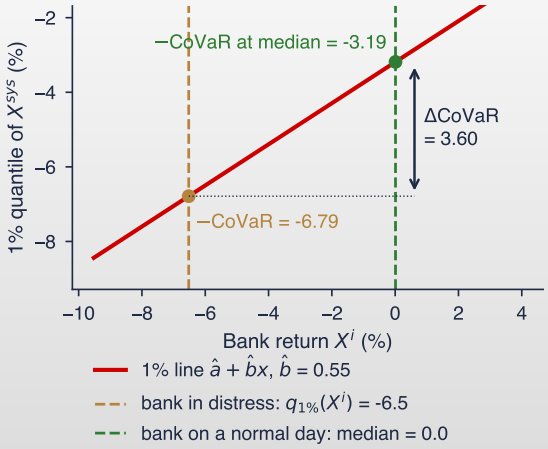
Prerequisites for Today (4/14): CoVaR and ΔCoVaR

- X^i: return of bank i; X^{sys}: return of the system; quantile regression of X^{sys} on X^i at level α gives $\hat{a}$, $\hat{b}$ Adrian & Brunnermeier (2016)
  - $q_\alpha(X^i)$, $q_{50\%}(X^i)$: the empirical α-quantile and median of the bank's returns
- CoVaR in distress, CoVaR at the median and their difference:

$$\mathrm{CoVaR}_\alpha = -(\hat{a} + \hat{b}\, q_\alpha(X^i)), \qquad \Delta\mathrm{CoVaR}_\alpha = \hat{b}\,(q_{50\%}(X^i) - q_\alpha(X^i))$$

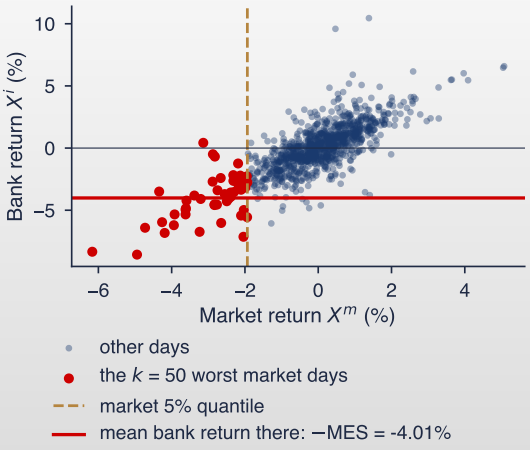
- Reading
  - CoVaR: the VaR of the system on a day when bank i is at its own VaR; at the median: on a normal day of the bank
  - ΔCoVaR: how much the VaR of the system rises when the bank moves from a normal day to distress, its contribution to systemic risk
  - large when the slope $\hat{b}$ is large and when the bank's own tail is far from its median

Prerequisites for Today (5/14): CoVaR on the 1% Line



- The 1% line of the chart in (3/14), read at two values of the bank return
  - at $q_{1\%}(X^i)$ (ochre): the bank is in distress; the height of the line is $-$CoVaR
  - at the median (green): a normal day of the bank
- ΔCoVaR is the vertical distance between the two points
  - it equals the slope $\hat{b}$ times the horizontal distance $q_{50\%} - q_{1\%}$
  - a bank with a large VaR but a small slope can have a small ΔCoVaR

Prerequisites for Today (6/14): Marginal Expected Shortfall



- **MES** Acharya, Pedersen, Philippon & Richardson (2017): the average loss of the bank on the market's worst α of days

$$\mathrm{MES}_\alpha = -E[X^i \mid X^m \le q_\alpha(X^m)]$$

- Notation
  - X^m: the market return; the bar reads "on the days when"; the minus sign makes the loss positive
- Estimate on n days
  - the $k = \lceil n\alpha \rceil$ days with the smallest market returns (red; $\lceil\cdot\rceil$: round up)
  - MES = − the mean bank return on those days; ES of the market = − the mean market return on the same days
  - MES/ES ≈ the tail beta of the bank: above 1, the bank falls more than the market

Prerequisites for Today (7/14): LRMES and SRISK

- **LRMES** Acharya, Engle & Richardson (2012): the loss of the bank if the market falls 40% in six months

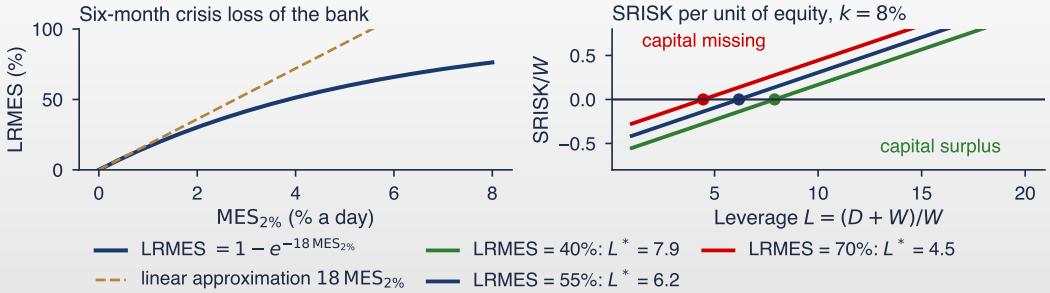
$$\mathrm{LRMES} = 1 - \exp(-18\,\mathrm{MES}_{\mathbf{2\%}})$$

- Notation
  - $\mathrm{MES}_{\mathbf{2\%}}$: the average loss of the bank (as a fraction) on the days when the market falls by more than 2%
  - the factor 18 turns this daily loss into a six-month crisis loss; LRMES lies between 0 and 1
- **SRISK** Brownlees & Engle (2017): the capital a bank would miss in a crisis

$$\mathrm{SRISK} = kD - (1-k)(1-\mathrm{LRMES})\,W$$

- Notation
  - k: the required capital ratio ($k = 8\%$); D: debt (book value); W: market value of equity
  - kD: capital required on the debt; $(1-k)(1-\mathrm{LRMES})W$: equity left after the crisis, net of the capital required on it
  - aggregate SRISK: the sum of the positive values; a negative SRISK is a capital surplus

Prerequisites for Today (8/14): Leverage and SRISK



- Left: LRMES is close to $18\,\mathrm{MES}_{2\%}$ for small MES, but bends below it and never exceeds 100%
- Right: with leverage $L = (D + W)/W$, $\mathrm{SRISK}/W = k(L - 1) - (1 - k)(1 - \mathrm{LRMES})$, a straight line in L
  - SRISK > 0 exactly when $L > L^* = 1 + (1 - k)(1 - \mathrm{LRMES})/k$ (the dots); a larger LRMES lowers L^*

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Prerequisites for Today (9/14): Correlation in Crises

- If $y = \beta x + e$ with a fixed β, the correlation rises when the variance of x rises Forbes & Rigobon (2002)
  - x: the source market; e: a shock independent of x
- **Forbes–Rigobon correction** of the crisis correlation ρ:

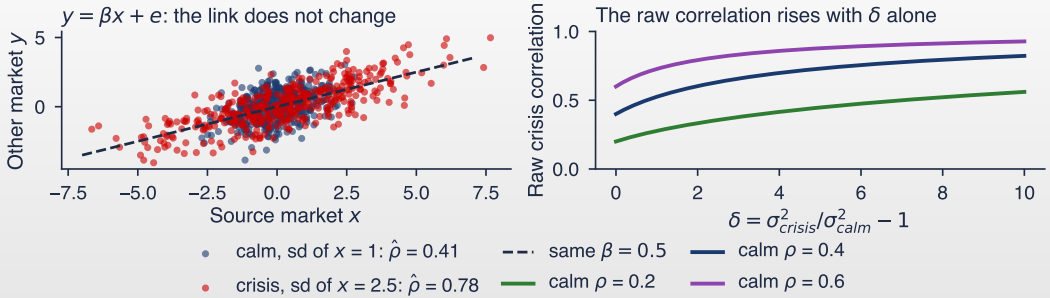
$$\rho^* = \frac{\rho}{\sqrt{1 + \delta(1 - \rho^2)}}, \qquad \delta = \frac{\sigma^2_{\text{crisis}}}{\sigma^2_{\text{calm}}} - 1$$

- Notation and reading
  - σ^2: the variance of the source market x; δ: its relative rise; ρ^*: the corrected correlation
  - **contagion**: the corrected correlation rises; **interdependence**: only the raw correlation rises
- **Fisher z test** of $H_0 : \rho_{\text{crisis}} = \rho_{\text{calm}}$ against a rise (independent samples of sizes n_1, n_0):

$$z = \frac{\operatorname{atanh} r_1 - \operatorname{atanh} r_0}{\sqrt{1/(n_1 - 3) + 1/(n_0 - 3)}}, \qquad \operatorname{atanh} r = \tfrac{1}{2} \ln \frac{1 + r}{1 - r}$$

- Reading
  - r_1, r_0: the crisis and calm correlations; atanh makes them approximately Normal; reject at 5% if $z > 1.645$ (one-sided test)

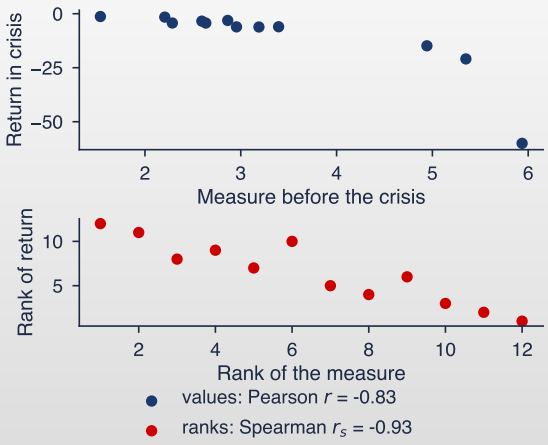
Prerequisites for Today (10/14): The Forbes–Rigobon Effect



- Left: the same relation $y = 0.5x + e$ in a calm period and in a crisis in which the standard deviation of x is 2.5 times larger: the raw correlation almost doubles
- Right: the raw crisis correlation $\rho^* \sqrt{1+\delta} / \sqrt{1 + \delta \rho^{*2}}$ for an unchanged link ρ^*: a rise in variance alone looks like contagion

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Prerequisites for Today (11/14): Rank Correlation and Crisis Returns

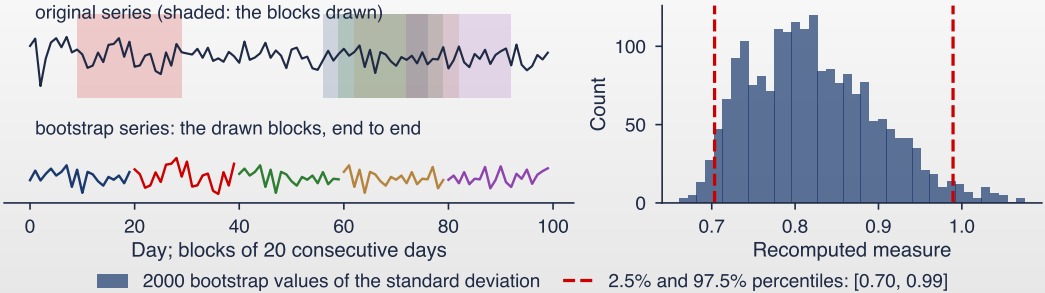


- **Spearman correlation**: the Pearson correlation of the ranks; without ties

$$r_s = 1 - \frac{6 \sum_{i=1}^{n} d_i^2}{n(n^2 - 1)}$$

- Notation and reading
  - d_i: the difference between the two ranks of unit i; n: the number of units (banks)
  - $r_s = 1$ (-1): the two orderings agree (are reversed); one outlier changes Pearson r much more
  - p-value from $t = r_s \sqrt{(n-2)/(1-r_s^2)} \sim t(n-2)$: with 11 banks only a strong link is significant
- Crisis return of a bank (buy and hold)
  - $100\,(P_{\text{end}}/P_{\text{start}} - 1)$: the simple return from the first to the last day of the episode

Prerequisites for Today (12/14): The Moving-Block Bootstrap



- **Moving-block bootstrap**: draw at random $\lceil n/b \rceil$ blocks of $b = 20$ consecutive days, join them, recompute the measure; repeat B times
  - blocks keep the volatility clustering of the returns (Chapter 8); the 2.5% and 97.5% percentiles of the B values give a 95% interval

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Prerequisites for Today (13/14): VAR Models and Granger Causality

- **VAR**(p) (vector autoregression) for the vector x_t of N returns:

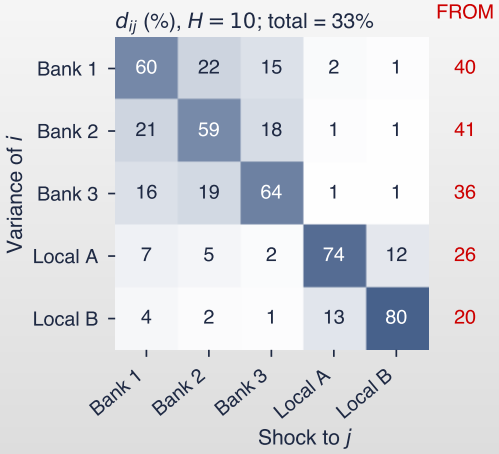
$$x_t = c + A_1 x_{t-1} + \cdots + A_p x_{t-p} + u_t, \qquad \mathrm{Var}(u_t) = \Sigma$$

- Notation
  - c: vector of constants; A_j: $N \times N$ matrices of coefficients at lag j; u_t: the shocks, with covariance matrix Σ
  - order p by **BIC** $= \ln|\hat{\Sigma}_p| + pN^2 \ln n/n$: fit against the number of coefficients; the smallest BIC wins
- **Granger causality** Granger (1969): in the equation of y, test that all lags of x have zero coefficients

$$y_t = c + \sum_{j=1}^{p} a_j y_{t-j} + \sum_{j=1}^{p} b_j x_{t-j} + e_t, \qquad H_0 : b_1 = \cdots = b_p = 0$$

- Reading
  - an F test; rejecting H_0: past values of x improve the forecast of y
  - a statistical link, not a proof of a causal channel

Prerequisites for Today (14/14): Connectedness



- **Connectedness** Diebold & Yilmaz (2012): from the VAR, the share of the H-day forecast error variance of i due to shocks to j
  - Θ_h: the response of x_{t+h} to u_t; e_i: the i-th unit vector
- Generalised decomposition (Pesaran–Shin):

$$d_{ij} \propto \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i^\top \Theta_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} e_i^\top \Theta_h \Sigma \Theta_h^\top e_i}$$

- Reading the table
  - rows rescaled to sum to 100%; σ_{jj}: the variance of shock j
  - FROM others $= \sum_{j \neq i} d_{ij}$; total $= \frac{1}{N} \sum_{i \neq j} d_{ij}$
  - chart: the two local banks take 20–26% of their risk from the other four banks

A1: CoVaR of the S&P 500 Given JPMorgan Chase [Solved]

Task

- Daily log returns (%) since 2005, $n = 5\,229$. Quantile regression at 1% of the S&P 500 on JPM: $\hat{a} = -2.35$, $\hat{b} = 0.39$; JPM: $q_{1\%} = -6.27$, $q_{50\%} = 0.07$; VaR 1% of the S&P 500: 3.53%.
- 1. Compute CoVaR 1% of the S&P 500 given that JPM is at its 1% quantile.
- 2. Compute CoVaR 1% at the median of JPM.
- 3. Compute ΔCoVaR 1%.
- 4. Compare CoVaR 1% with the unconditional VaR 1% of the S&P 500.
- Report: four numbers and one sentence.

Solution

- 1. $-(-2.35 + 0.39 \times (-6.27)) = 2.35 + 2.4453 = 4.7953 \approx 4.80\%$
- 2. $-(-2.35 + 0.39 \times 0.07) = 2.35 - 0.0273 = 2.3227 \approx 2.32\%$
- 3. $0.39 \times (0.07 - (-6.27)) = 0.39 \times 6.34 = 2.4726 \approx 2.47\%$ (the difference of 1 and 2)
- 4. $4.80/3.53 = 1.36$
- When JPM is in distress, the VaR 1% of the index is about 1.36 times its unconditional value.

Back Exercise list

A2: Bank of America [Proposed]

Solution discussed in class

Task

- Same data as A1. Quantile regression at 1% of the S&P 500 on BAC: $\hat{a} = -2.65$, $\hat{b} = 0.26$; BAC: $q_{1\%} = -8.13$, $q_{50\%} = 0.06$. Model: A1.
- 1. Compute VaR 1% of BAC, CoVaR 1% and CoVaR 1% at the median.
- 2. Compute ΔCoVaR 1%.
- 3. Rank JPM and BAC by VaR 1% and by ΔCoVaR 1%.
- Report: four numbers and one sentence.

Data for A3 and A4: 20 Days of March 2020

Day	S&P 500	JPM	GS	Day	S&P 500	JPM	GS
2.03	4.50	4.55	4.24	16.03	−12.77	−16.21	−13.59
3.03	−2.85	−3.82	−2.93	17.03	5.82	5.93	2.56
4.03	4.13	2.44	2.58	18.03	−5.32	−11.12	−12.50
5.03	−3.45	−5.03	−4.88	19.03	0.47	1.67	6.54
6.03	−1.72	−5.31	−3.03	20.03	−4.43	−2.13	−7.70
9.03	−7.90	−14.56	−10.97	23.03	−2.97	−5.50	−2.52
10.03	4.82	7.48	6.46	24.03	8.97	11.24	12.93
11.03	−5.01	−4.82	−7.00	25.03	1.15	3.66	0.99
12.03	−9.99	−8.60	−13.17	26.03	6.05	6.73	6.65
13.03	8.88	16.56	16.20	27.03	−3.43	−7.39	−4.60

- Daily log returns in %, 2–27 March 2020 (day.month); JPM: JPMorgan Chase, GS: Goldman Sachs
- The same table is in the notebook (section A3)

A3: MES and LRMES of JPMorgan Chase [Solved]

Task

- Use the 20 days of the table (market: S&P 500).
- 1. Find the $k = \lceil 20 \times 0.10 \rceil$ worst market days and compute MES 10% of JPM.
- 2. Compute ES 10% of the market on the same days and the ratio MES/ES.
- 3. Compute $\mathrm{MES}_{2\%}$: the average loss of JPM on the days when the S&P 500 fell by more than 2%.
- 4. Compute LRMES.
- Report: five numbers and one sentence.

Solution

- 1. $k = 2$: 16.03 (-12.77) and 12.03 (-9.99); MES 10% $= -(-16.21 + (-8.60))/2 = 12.405\%$
- 2. ES 10% $= 11.38\%$; MES/ES $= 1.09$
- 3. 10 days below -2%; the JPM returns on them sum to -79.18: $\mathrm{MES}_{2\%} = 7.918\%$
- 4. $1 - \exp(-18 \times 7.918/100) = 1 - \exp(-1.4252) = 1 - 0.2405 = 75.95\%$
- On the market's worst days JPM lost more than the market; measured in this crisis month, LRMES is very high.

Back Exercise list

A4: MES and LRMES of Goldman Sachs [Proposed]

Solution discussed in class

Task

- Use the same table for GS. Model: A3.
- 1. Compute MES 10% of GS and the ratio MES/ES of the market.
- 2. Compute $\mathrm{MES}_{2\%}$ and LRMES of GS.
- 3. Compare GS with JPM on both measures.
- Report: four numbers and one sentence.

Back Exercise list

A5: SRISK of One Bank [Solved]

Task

- A bank has debt $D = 900$ and market value of equity $W = 100$ (billion RON); LRMES $= 55\%$; $k = 8\%$.
- 1. Compute the leverage L and SRISK.
- 2. Compute the break-even leverage L^*.
- 3. Recompute L and SRISK after a fall of 30% in W, with D unchanged.
- Report: five numbers and one sentence.

Solution

- 1. $L = 1000/100 = 10$; SRISK $= 0.08 \times 900 - 0.92 \times 0.45 \times 100 = 72.00 - 41.40 = 30.60$ billion
- 2. $L^* = 1 + 0.92 \times 0.45/0.08 = 6.175$; $L = 10 > L^*$, so SRISK > 0
- 3. $W = 70$: $L = 970/70 = 13.86$; SRISK $= 72.00 - 28.98 = 43.02$ billion
- A fall in the share price raises leverage and SRISK at the same time: SRISK grows in a crisis before any loan defaults.

Back Exercise list

A6: SRISK of Two Banks [Proposed]

Solution discussed in class

Task

- Bank X: $D = 570$, $W = 30$, LRMES $= 50\%$; bank Y: $D = 180$, $W = 60$, LRMES $= 65\%$ (billion RON). Model: A5.
- 1. Compute the leverage and SRISK of each bank with $k = 8\%$.
- 2. Compute the aggregate SRISK and the share of each bank.
- 3. Recompute both SRISK values with $k = 5.5\%$.
- Report: six numbers and one sentence.

Back Exercise list

B1: MES and ΔCoVaR of the US Banks [Solved]

- **Question**: do MES and ΔCoVaR point to the same most systemic US bank?
- **Inputs**: six US banks and the S&P 500, daily log returns since 2005
- **Tasks**
  1. Compute MES 5% of each bank with the S&P 500 as the market.
  2. Compute ΔCoVaR 1% of the S&P 500 given each bank by quantile regression.
  3. Compute 95% moving-block bootstrap intervals for both measures (blocks of 20 days).
  4. Compute the Spearman correlation between the two rankings.
  5. Interpretation: do the two measures agree on the most systemic bank?
- **Report**: a table of twelve numbers with intervals, the chart and two sentences
- Notebook: section B1

Back Exercise list

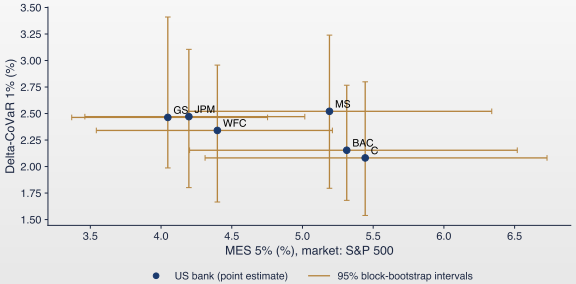
B1: Solution (1/2) [Solved]

	MES 5%	95% interval	ΔCoVaR 1%	95% interval
JPM	4.20	[3.46, 5.02]	2.47	[1.80, 3.11]
BAC	5.31	[4.20, 6.52]	2.15	[1.68, 2.77]
C	5.44	[4.31, 6.73]	2.08	[1.54, 2.80]
GS	4.05	[3.37, 4.75]	2.46	[1.99, 3.41]
MS	5.19	[4.19, 6.34]	2.52	[1.80, 3.24]
WFC	4.40	[3.54, 5.21]	2.34	[1.67, 2.96]

- Market: the S&P 500; intervals: moving-block bootstrap, blocks of 20 days
- MES is about twice as large as ΔCoVaR: the two measures are on different scales and answer different questions

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B1: Solution (2/2) [Solved]



- $n = 5\,229$ days; largest MES: Citigroup; largest ΔCoVaR: Morgan Stanley; Spearman correlation of the rankings -0.60
- Interpretation: no: MES ranks first the banks that fall most with the market (Citigroup, Bank of America), ΔCoVaR the banks whose distress moves the market most; the intervals overlap, so neither ranking is sharp

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B2: European and Romanian Banks [Proposed]

- **Question**: does the disagreement between MES and ΔCoVaR found in B1 also appear in Europe and in Romania? Model: B1.
- **Inputs**: five European banks and the Euro Stoxx 50 since 2005; TLV, BRD and the BET since 2010
- **Tasks**
  1. Compute MES 5% and ΔCoVaR 1% with bootstrap intervals for each bank, against its own market.
  2. Compute the Spearman correlation of the two rankings for the European banks.
  3. Compare the two Romanian banks on both measures.
  4. Interpretation: is the disagreement between MES and ΔCoVaR as strong in Europe as in the US?
- **Report**: one table and two sentences
- Notebook: section B2

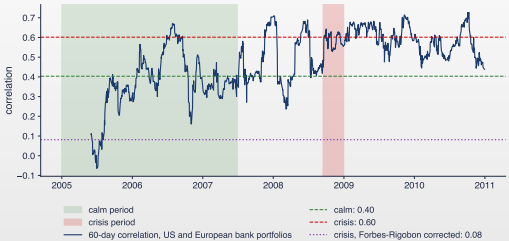
Back Exercise list

B3: US and European Banks in 2008 [Solved]

- **Question**: did the link between US and European banks become stronger after Lehman: contagion or interdependence?
- **Inputs**: equal-weighted portfolios of the six US and the five European banks, common days; calm: January 2005 – June 2007; crisis: 15 September – 31 December 2008
- **Tasks**
  1. Compute the correlation of the two portfolios in the calm and in the crisis period.
  2. Test the rise in correlation with the one-sided Fisher z test at 5%.
  3. Compute δ from the variance of the US portfolio and the Forbes–Rigobon corrected correlation.
  4. Repeat the test with the corrected correlation.
  5. Interpretation: contagion or interdependence?
- **Report**: six numbers, two decisions, the chart and two sentences
- Notebook: section B3

Back Exercise list

B3: Solution [Solved]



- $r_0 = 0.40$ ($n_0 = 506$), $r_1 = 0.60$ ($n_1 = 72$); $z = (0.695 - 0.428)/0.128 = 2.09$, p $= 0.018$: the raw rise is significant
- Variance of the US portfolio: $0.89 \to 75.39$, $\delta = 83.68$; $\rho^* = 0.60/\sqrt{1 + 83.68 \times 0.638} = 0.08$; $z = -2.69$, p $= 0.996$: no rise
- Interpretation: interdependence: the correlation rose because the variance of the US banks rose about 85 times; after the correction the link is not stronger Forbes & Rigobon (2002)

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B4: European and Romanian Banks in 2020 [Proposed]

- **Question**: did the link between European and Romanian banks become stronger in the COVID-19 crash? Model: B3.
- **Inputs**: equal-weighted portfolios of the five European banks and of TLV and BRD, common days since 2010; calm: 2019; crisis: 20 February – 30 April 2020
- **Tasks**
  1. Compute the correlation in the calm and in the crisis period and test the rise with the Fisher z test.
  2. Compute δ from the variance of the European portfolio and the corrected correlation.
  3. Repeat the test with the corrected correlation.
  4. Interpretation: contagion or interdependence?
- **Report**: six numbers, two decisions and two sentences
- Notebook: section B4

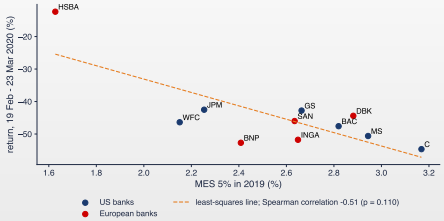
Back Exercise list

B5: Did MES in 2019 Rank the Losses of March 2020? [Solved]

- **Question**: do the banks with a high MES before a crisis lose more in it, as in Acharya, Pedersen, Philippon & Richardson (2017)?
- **Inputs**: 11 US and European banks; MES 5% on the daily returns of 2019; buy-and-hold return from 19 February to 23 March 2020
- **Tasks**
  1. Compute MES 5% of each bank in 2019 against its own market.
  2. Compute the return of each bank from 19 February to 23 March 2020.
  3. Compute the Spearman correlation between MES and the return and its p-value.
  4. Interpretation: did MES rank the March 2020 losses?
- **Report**: the chart, the correlation with its p-value and two sentences
- Notebook: section B5

Back Exercise list

B5: Solution [Solved]



- Highest MES in 2019: Citigroup (3.17%), also the worst return (-54.7%); lowest MES: HSBC (1.63%), also the smallest loss (-12.3%)
- Spearman correlation -0.51, $p = 0.110$ ($n = 11$)
- Interpretation: partly: the sign is the one of Acharya, Pedersen, Philippon & Richardson (2017) and the extremes match, but with 11 banks the correlation is not significant at 5%; the middle of the ranking is noise

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B6: Did MES in 2022 Rank the Losses of March 2023? [Proposed]

- **Question**: does the result of B5 hold in the banking turmoil of March 2023? Model: B5.
- **Inputs**: 13 banks (with TLV and BRD); MES 5% on the daily returns of 2022; return from 28 February to 31 March 2023
- **Tasks**
  1. Compute MES 5% of each bank in 2022 against its own market.
  2. Compute the return of each bank from 28 February to 31 March 2023.
  3. Compute the Spearman correlation and its p-value.
  4. Interpretation: why may MES fail to rank the losses of March 2023?
- **Report**: one table, the correlation with its p-value and two sentences
- Notebook: section B6

Back Exercise list

C1: How Connected Are Romanian and Euro-Area Banks? [Proposed]

- **Question**: what share of the risk of TLV and BRD comes from the European banks, and does it rise in crises?
- **Inputs**: the five European banks, TLV and BRD, daily log returns on common days since 2010; models: B1, the connectedness slide
- **Tasks**
  1. Fit a VAR to the seven return series (order by BIC) and compute the connectedness table for $H = 10$ days.
  2. Compute the share of the forecast error variance of TLV and BRD that comes from the five European banks.
  3. Repeat it on rolling windows of 250 days and compare 2019 with 2020.
  4. Interpretation: is the systemic risk of the Romanian banks imported?
- **Report**: a table and a plan for a project
- Notebook: section C1

C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant to summarise the systemic risk of JPMorgan Chase. The answer:
- (a) The CoVaR 99% of the S&P 500 given JPM is −4.80%.
- (b) Delta-CoVaR = CoVaR - VaR of JPM = 4.80 - 6.27 = −1.48%, so JPM reduces systemic risk.
- (c) MES 5% is the average loss of the S&P 500 on the worst 5% of days of JPM.
- (d) The bank with the largest VaR 1% is always the most systemic one.
- (e) The correlation of US and European banks rose from 0.40 to 0.60 in autumn 2008, which proves contagion.
- (f) A positive SRISK means that the bank has more capital than it needs in a crisis.
- (g) Granger causality from bank A to bank B means that past returns of A help to forecast B; it does not prove a causal channel.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - 2. Report: a list of seven verdicts with one line of justification each.

Points to Remember

- CoVaR 1% $= -(\hat{a} + \hat{b}\, q_{1\%}(X^i))$ and $\Delta\text{CoVaR} = \hat{b}\,(q_{50\%} - q_{1\%})$: the system given the bank
- MES: the bank given the market; SRISK $= kD - (1-k)(1 - \text{LRMES})W$ grows with leverage
- VaR, MES and ΔCoVaR rank banks differently; rankings need bootstrap intervals
- A rise in correlation in a crisis is not proof of contagion: correct for volatility (Forbes–Rigobon)
- An AI answer is a draft: check the level, the sign, the direction of conditioning and the definition of ΔCoVaR
- Next
  - Lecture 15 develops these topics and adds the crises, CoVaR with state variables, networks and macroprudential policy
  - C1 can grow into a team project: Romanian and euro-area banks, connectedness and ΔCoVaR in 2011–2012, 2020 and 2023
  - Reading: Adrian & Brunnermeier (2016); Acharya, Pedersen, Philippon & Richardson (2017); Brownlees & Engle (2017); survey: Benoit, Colliard, Hurlin & Pérignon (2017)

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